

Lingofy

Music-Driven Language Learning through Human-Computer Interaction

Research Project Report

Authors

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An HCI research prototype exploring how music-driven interaction, personalization, and engagement influence language learning outcomes.

Repo: <https://github.com/Unnati1007/Lingofy.git>

Stack: React 19 + TypeScript + Vite (frontend) · Node.js + Express + TypeScript + MongoDB (backend) · Groq LLaMA 3.3 (AI)

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1. Abstract

Learning a new language can be challenging because many learners lose motivation with traditional text-based methods. **Lingofy** is an AI-powered language learning platform that uses music to make learning more engaging and interactive. It combines synchronized song lyrics, AI-generated vocabulary quizzes, pronunciation practice, and personalized learning paths to help users learn languages naturally.

The project investigates whether learning through music improves learner engagement, vocabulary retention, and overall learning performance compared to traditional learning methods. Two learning modes—**Music Mode** and **Traditional Mode**—are used to compare users' learning outcomes and experiences. Through this research, Lingofy aims to contribute to Human–Computer Interaction (HCI) and AI-assisted education by exploring how music and personalization can create a more effective and enjoyable language learning experience.

Keywords: *Human–Computer Interaction (HCI), Artificial Intelligence, Language Learning, Music-Based Learning, Personalized Learning, Educational Technology.*

2. Introduction

Language learning applications such as Duolingo and Memrise have made language education more accessible, but maintaining learner motivation and long-term engagement remains a challenge. Most existing platforms rely on repetitive text-based exercises that often lack real-world context and interactive learning experiences.

Music has been shown to improve memory, pronunciation, and emotional engagement, making it a promising tool for language acquisition. However, few language learning systems integrate music as the primary learning medium while also providing personalized AI-assisted learning.

Lingofy addresses this gap by combining music, artificial intelligence, and Human–Computer Interaction (HCI) principles into a single learning platform. It allows users to learn through songs using synchronized lyrics, AI-generated quizzes, pronunciation practice, and personalized learning pathways. The project also evaluates whether music-based learning improves engagement and learning outcomes compared to traditional text-based methods.

3. Research Objectives

The primary objective of Lingofy is to develop and evaluate an AI-powered music-based language learning platform that enhances learner engagement and vocabulary retention.

The project specifically aims to:

- Develop an interactive language-learning platform that uses music and AI.
- Compare music-based learning with traditional text-based learning.
- Evaluate learner engagement, motivation, and learning performance.
- Provide personalized learning experiences based on user progress.
- Contribute to Human–Computer Interaction (HCI) research in educational technology.

Research Questions

- Does music-based learning improve vocabulary retention?
 - Does music increase learner engagement and motivation?
 - Can AI-driven personalization improve learning outcomes?
 - How does music-based learning influence the overall user experience?
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4. Proposed Solution: Lingofy

Lingofy is an AI-powered language learning platform that transforms music into an interactive educational experience. Instead of relying only on traditional vocabulary exercises, learners study languages through songs while interacting with synchronized lyrics, AI-generated quizzes, pronunciation practice, and personalized learning recommendations.

The learning process begins with song selection, followed by synchronized lyric presentation and AI-based vocabulary extraction. The system then generates contextual quizzes, evaluates learner performance, and provides adaptive recommendations based on progress. This learner-centered approach combines authentic language exposure with intelligent personalization to create an engaging and effective learning environment.

Key Features

- Interactive music player with synchronized lyrics
- AI-generated vocabulary and grammar quizzes
- Pronunciation assessment
- Personalized learning pathways
- Learning analytics and progress tracking
- Achievement and reward system
- Music Mode and Traditional Mode for comparative research

By integrating music, AI, and HCI principles, Lingofy aims to create a language learning experience that is both educationally effective and enjoyable.

5. System Architecture

Lingofy follows a modular client-server architecture that integrates music-based learning, artificial intelligence, learning analytics, and research evaluation within a unified platform. The architecture consists of seven major components: the Client Layer, User Interface Module, Backend Services, Database Layer, Data Collection Layer, Research Analytics Layer, and Ethics & Privacy Layer. Together, these components support personalized language learning while enabling controlled HCI experimentation.

Figure 1. Overall System Architecture of Lingofy

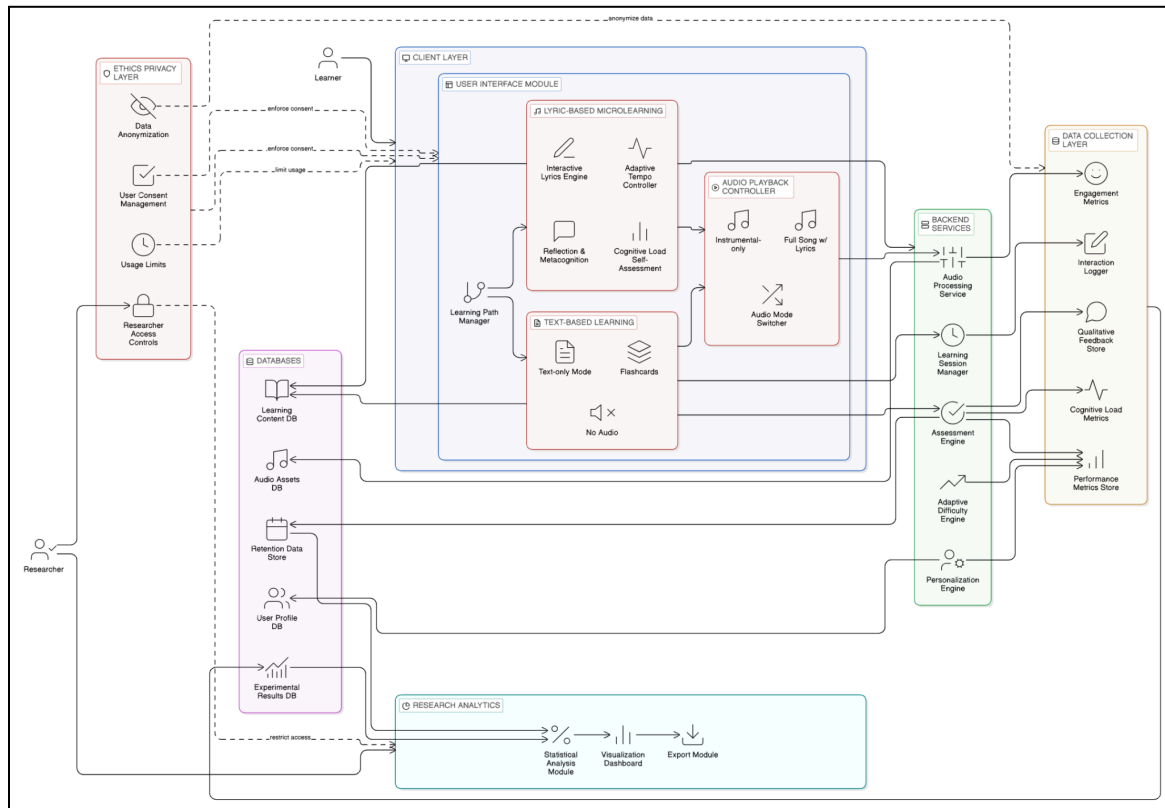


Figure 1 illustrates the overall architecture of Lingofy. The system integrates lyric-based learning, traditional text-based learning, AI-powered personalization, backend processing, research analytics, and privacy-preserving data collection within a unified human–computer interaction framework.

The client layer provides an interactive learning interface where users access both lyric-based and traditional learning modes. Backend services manage authentication, AI-powered quiz generation, personalization, adaptive difficulty, and learning sessions. User profiles, learning content, audio assets, and experimental results are securely stored in dedicated databases.

The platform also includes a dedicated data collection layer that records engagement, interaction logs, cognitive load, qualitative feedback, and learning performance. These anonymized data are analyzed through the Research Analytics Layer to evaluate learner behavior and compare outcomes between Music Mode and Traditional Mode. An Ethics & Privacy Layer ensures user consent, data anonymization, access control, and responsible use of research data.

6. Research Methodology

This study follows a between-subjects experimental design to compare music-based learning with traditional text-based learning.

Participants will be randomly assigned to one of two groups:

Group	Learning Method
Group A (Experimental)	Music-Based Learning using Lingofy
Group B (Control)	Traditional Text-Based Learning

Participants will complete learning sessions followed by vocabulary quizzes, pronunciation exercises, and feedback questionnaires. The study will evaluate learning effectiveness using quiz performance, vocabulary retention, user engagement, session duration, cognitive load, and overall user satisfaction.

All participant data will be collected with informed consent, anonymized before analysis, and used solely for academic research.

6.4 Experimental Procedure

The study consists of the following stages:

1. Participants register and provide informed consent.
 2. A pre-test is conducted to assess baseline language proficiency.
 3. Participants are randomly assigned to either the Music Mode or Traditional Mode.
 4. Each participant completes multiple learning sessions using the assigned learning mode.
 5. After each session, participants complete quizzes and questionnaires.
 6. A post-test is conducted to measure learning improvement.
 7. Learning analytics and participant feedback are collected for analysis.
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7. Implementation Overview

Lingofy is developed as a full-stack web application that combines Artificial Intelligence with modern web technologies to deliver an interactive language learning experience. The frontend is built using **React** and **TypeScript**, while the backend uses **Node.js**, **Express.js**, and **MongoDB**. AI is integrated to generate contextual quizzes, personalize learning activities, and adapt lesson difficulty according to learner performance.

The platform currently supports user authentication, music-based learning, interactive lyrics, AI quiz generation, pronunciation assessment, learning analytics, and personalized learning paths. The core system has been implemented successfully, while the experimental user study and evaluation are currently in progress.

8. Expected Outcomes and Contributions

Lingofy is expected to demonstrate that integrating music with AI-assisted learning can improve learner engagement, motivation, and vocabulary retention compared to traditional language learning methods. The project aims to contribute to the fields of Human–Computer Interaction, Educational Technology, and Artificial Intelligence by providing a personalized and interactive learning platform. The research findings may also support the future development of intelligent educational systems that combine music, adaptive learning, and user-centered design to create more effective language learning experiences.

9. Conclusion

Lingofy presents a novel approach to language learning by integrating music, Artificial Intelligence, and Human–Computer Interaction into a unified educational platform. Through its comparative evaluation of music-based and traditional learning methods, the project aims to explore how interactive and personalized learning experiences influence learner engagement and language acquisition. The outcomes of this research are expected to contribute to the design of future AI-powered educational technologies and support further research in music-assisted language learning.
